

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

file_path = "/content/superstore.csv.csv"

try:
    df = pd.read_csv(file_path, encoding="latin1")
except FileNotFoundError:
    print("ERROR: superstore.csv was not found.")
    print("Place the CSV file in the same folder as this Python file.")
    exit()

print("\n===== DATASET LOADED SUCCESSFULLY =====")

print("\nFirst 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

df.columns = df.columns.str.strip()

required_columns = [
    "Sales",
    "Profit",
    "Category",
    "Sub-Category",
    "Region"
]

missing_columns = [
    col for col in required_columns if col not in df.columns
]

if missing_columns:
    print("\nMissing required columns:", missing_columns)
    print("Available columns:", df.columns.tolist())
    exit()

df["Sales"] = pd.to_numeric(df["Sales"], errors="coerce")
df["Profit"] = pd.to_numeric(df["Profit"], errors="coerce")

df = df.dropna(subset=["Sales", "Profit"])

total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique() if "Order ID" in df.columns else len(df)

profit_margin = (total_profit / total_sales) * 100

print("\n===== KEY PERFORMANCE INDICATORS =====")

print(f"Total Sales      : ${total_sales:,.2f}")
print(f"Total Profit     : ${total_profit:,.2f}")
print(f"Total Orders      : {total_orders:,}")
print(f"Profit Margin     : {profit_margin:.2f}%")

category_sales = (
    df.groupby("Category")["Sales"]
    .sum()
    .sort_values(ascending=False)

```

```

)

print("\n===== SALES BY CATEGORY =====")
print(category_sales)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=category_sales.index,
    y=category_sales.values
)

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()

category_profit = (
    df.groupby("Category")["Profit"]
    .sum()
    .sort_values(ascending=False)
)

print("\n===== PROFIT BY CATEGORY =====")
print(category_profit)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=category_profit.index,
    y=category_profit.values
)

plt.title("Profit by Category")
plt.xlabel("Category")
plt.ylabel("Profit")

plt.tight_layout()
plt.show()

region_sales = (
    df.groupby("Region")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

print("\n===== SALES BY REGION =====")
print(region_sales)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=region_sales.index,
    y=region_sales.values
)

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.xticks(rotation=30)

plt.tight_layout()
plt.show()

region_profit = (
    df.groupby("Region")["Profit"]
    .sum()
    .sort_values(ascending=False)
)

print("\n===== PROFIT BY REGION =====")
print(region_profit)

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plt.figure(figsize=(8, 5))

sns.barplot(
    x=region_profit.index,
    y=region_profit.values
)

plt.title("Profit by Region")
plt.xlabel("Region")
plt.ylabel("Profit")

plt.xticks(rotation=30)

plt.tight_layout()
plt.show()

subcategory_sales = (
    df.groupby("Sub-Category")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

print("\n===== TOP SUB-CATEGORIES BY SALES =====")
print(subcategory_sales.head(10))

plt.figure(figsize=(10, 6))

subcategory_sales.head(10).sort_values().plot(
    kind="barh"
)

plt.title("Top 10 Sub-Categories by Sales")
plt.xlabel("Sales")
plt.ylabel("Sub-Category")

plt.tight_layout()
plt.show()

subcategory_profit = (
    df.groupby("Sub-Category")["Profit"]
    .sum()
    .sort_values(ascending=False)
)

print("\n===== TOP SUB-CATEGORIES BY PROFIT =====")
print(subcategory_profit.head(10))

plt.figure(figsize=(10, 6))

subcategory_profit.head(10).sort_values().plot(
    kind="barh"
)

plt.title("Top 10 Sub-Categories by Profit")
plt.xlabel("Profit")
plt.ylabel("Sub-Category")

plt.tight_layout()
plt.show()

loss_categories = subcategory_profit[subcategory_profit < 0]

print("\n===== LOSS-MAKING SUB-CATEGORIES =====")
print(loss_categories)

correlation = df[["Sales", "Profit"]].corr()

print("\n===== CORRELATION =====")
print(correlation)

plt.figure(figsize=(6, 5))

sns.heatmap(

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```

        correlation,
        annot=True,
        cmap="coolwarm",
        fmt=".2f"
    )

plt.title("Sales-Profit Correlation")

plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))

sns.histplot(
    df["Sales"],
    bins=30,
    kde=True
)

plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))

sns.histplot(
    df["Profit"],
    bins=30,
    kde=True
)

plt.title("Profit Distribution")
plt.xlabel("Profit")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category",
    alpha=0.6
)

plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.tight_layout()
plt.show()

fig = px.bar(
    category_sales.reset_index(),
    x="Category",
    y="Sales",
    title="Interactive Sales by Category",
    text_auto=".2s"
)

fig.show()

fig = px.bar(
    region_sales.reset_index(),
    x="Region",
    y="Sales",
    title="Interactive Sales by Region",
    text_auto=".2s"
)

fig.show()

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fig = px.scatter(
    df,
    x="Sales",
    y="Profit",
    color="Category",
    hover_data=["Sub-Category"],
    title="Interactive Sales vs Profit"
)

fig.show()

if "Product Name" in df.columns:

    product_sales = (
        df.groupby("Product Name")["Sales"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
    )

    print("\n===== TOP 10 PRODUCTS BY SALES =====")
    print(product_sales)

print("\n")
print("=" * 60)
print("                BUSINESS INSIGHTS")
print("=" * 60)

best_category = category_sales.idxmax()
best_region = region_sales.idxmax()
best_profit_category = category_profit.idxmax()

print(f"\n1. The category with the highest sales is: {best_category}")

print(f"\n2. The region with the highest sales is: {best_region}")

print(
    f"\n3. The category generating the highest total profit is: "
    f"{best_profit_category}"
)

if len(loss_categories) > 0:
    print(
        "4. Loss-making sub-categories were identified and "
        "should be investigated for pricing and cost issues."
    )
else:
    print(
        "4. No loss-making sub-category was identified "
        "in the dataset."
    )

print(
    f"\n5. Overall profit margin is approximately "
    f"{profit_margin:.2f}%"
)

print(
    "6. Sales and profit visualizations can help management "
    "identify profitable categories and regions."
)

print("\n")
print("=" * 60)
print("                RECOMMENDATIONS")
print("=" * 60)

print("""
1. Focus on high-performing product categories.

2. Investigate low-profit or loss-making sub-categories.

3. Analyze regional performance before allocating resources.

4. Improve pricing and discount strategies for low-profit products.

```

5. Monitor Sales and Profit KPIs regularly.

6. Use interactive dashboards to communicate business performance to stakeholders.

7. Continue analyzing customer and product-level trends to identify new opportunities.

""")

print("\n===== ANALYSIS COMPLETED =====")

===== DATASET LOADED SUCCESSFULLY =====

First 5 rows:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode
0	40098	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class
1	26341	IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class
2	25330	IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class
3	13524	ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class
4	47221	SG-2014-RH9495111-41948	11/5/2014	11/6/2014	Same Day

	Customer ID	Customer Name	Segment	Postal Code	City
0	AB-100151402	Aaron Bergman	Consumer	73120.0	Oklahoma City
1	JR-162107	Justin Ritter	Corporate	NaN	Wollongong
2	CR-127307	Craig Reiter	Consumer	NaN	Brisbane
3	KM-1637548	Katherine Murray	Home Office	NaN	Berlin
4	RH-9495111	Rick Hansen	Consumer	NaN	Dakar

	Product ID	Category	Sub-Category
0	TEC-PH-5816	Technology	Phones
1	FUR-CH-5379	Furniture	Chairs
2	TEC-PH-5356	Technology	Phones
3	TEC-PH-5267	Technology	Phones
4	TEC-CO-6011	Technology	Copiers

	Product Name	Sales	Quantity	Discount
0	Samsung Convoy 3	221.98	2	0.0
1	Novimex Executive Leather Armchair, Black	3709.40	9	0.1
2	Nokia Smart Phone, with Caller ID	5175.17	9	0.1
3	Motorola Smart Phone, Cordless	2892.51	5	0.1
4	Sharp Wireless Fax, High-Speed	2832.96	8	0.0

	Profit	Shipping Cost	Order Priority
0	62.15	40.77	High
1	-288.77	923.63	Critical
2	919.97	915.49	Medium
3	-96.54	910.16	Medium
4	311.52	903.04	Critical

[5 rows x 24 columns]

Dataset Shape:
(1000, 24)

Column Names:

['i»¿Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Postal Code', 'C

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1000 entries, 0 to 999

Data columns (total 24 columns):

#	Column	Non-Null Count	Dtype
0	i»¿Row ID	1000 non-null	int64
1	Order ID	1000 non-null	object
2	Order Date	1000 non-null	object
3	Ship Date	1000 non-null	object
4	Ship Mode	1000 non-null	object
5	Customer ID	1000 non-null	object
6	Customer Name	1000 non-null	object
7	Segment	1000 non-null	object
8	Postal Code	194 non-null	float64
9	City	1000 non-null	object
10	State	1000 non-null	object
11	Country	1000 non-null	object
12	Region	1000 non-null	object
13	Market	1000 non-null	object
14	Product ID	1000 non-null	object
15	Category	1000 non-null	object
16	Sub-Category	1000 non-null	object
17	Product Name	1000 non-null	object
18	Sales	1000 non-null	float64
19	Quantity	1000 non-null	int64
20	Discount	1000 non-null	float64
21	Profit	1000 non-null	float64
22	Shipping Cost	1000 non-null	float64
23	Order Priority	1000 non-null	object

dtypes: float64(5), int64(2), object(17)

memory usage: 187.6+ KB

None

Missing Values:

i»¿Row ID	0
Order ID	0

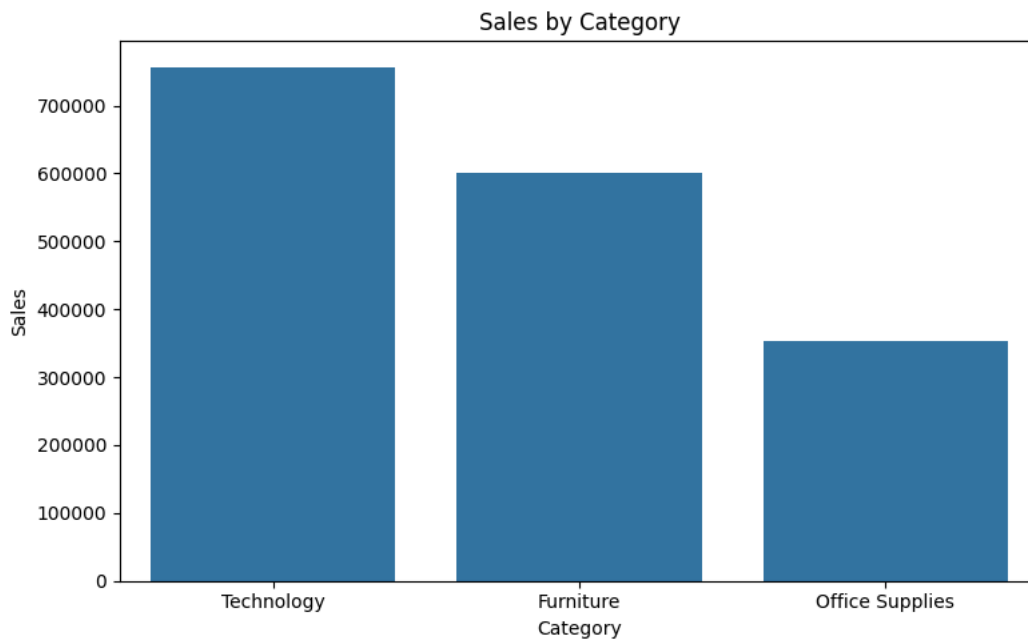
```
order date      0
Ship Date      0
Ship Mode      0
Customer ID    0
Customer Name   0
Segment       0
Postal Code    806
City           0
State          0
Country        0
Region         0
Market         0
Product ID     0
Category       0
Sub-Category   0
Product Name   0
Sales          0
Quantity       0
Discount       0
Profit         0
Shipping Cost  0
Order Priority  0
dtype: int64
```

===== KEY PERFORMANCE INDICATORS =====

```
Total Sales      : $1,710,971.47
Total Profit     : $288,920.44
Total Orders     : 867
Profit Margin    : 16.89%
```

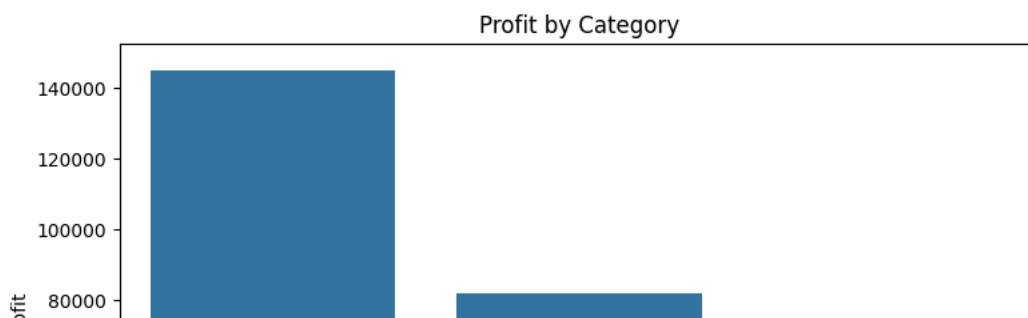
===== SALES BY CATEGORY =====

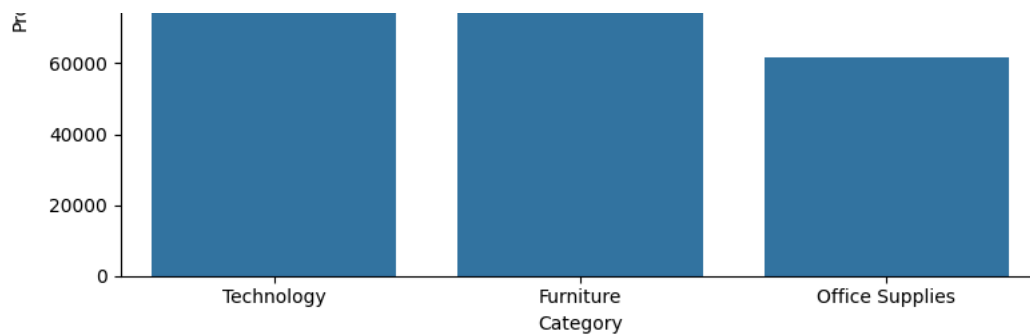
```
Category
Technology      756983.02
Furniture       601492.31
Office Supplies  352496.14
Name: Sales, dtype: float64
```



===== PROFIT BY CATEGORY =====

```
Category
Technology      145068.68
Furniture       82188.95
Office Supplies  61662.81
Name: Profit, dtype: float64
```

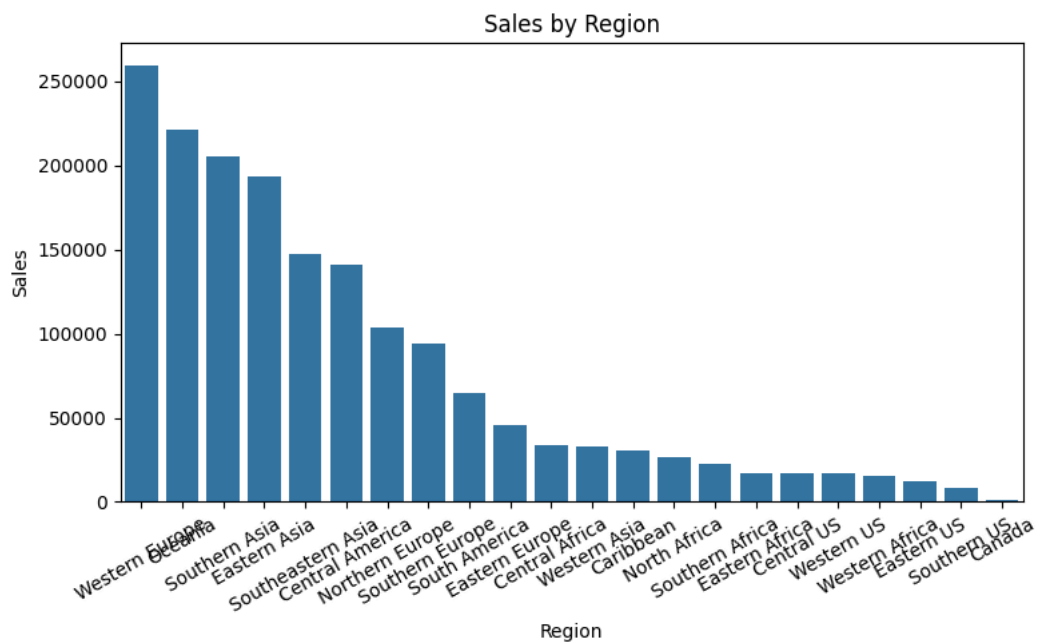




===== SALES BY REGION =====

Region	Sales
Western Europe	259576.28
Oceania	220809.08
Southern Asia	205466.26
Eastern Asia	193590.60
Southeastern Asia	147337.60
Central America	140966.21
Northern Europe	103888.63
Southern Europe	94441.11
South America	64550.54
Eastern Europe	45662.19
Central Africa	33810.48
Western Asia	33308.21
Caribbean	30746.74
North Africa	26266.44
Southern Africa	22262.28
Eastern Africa	17387.94
Central US	17187.94
Western US	16729.95
Western Africa	15478.79
Eastern US	12256.44
Southern US	8360.75
Canada	887.01

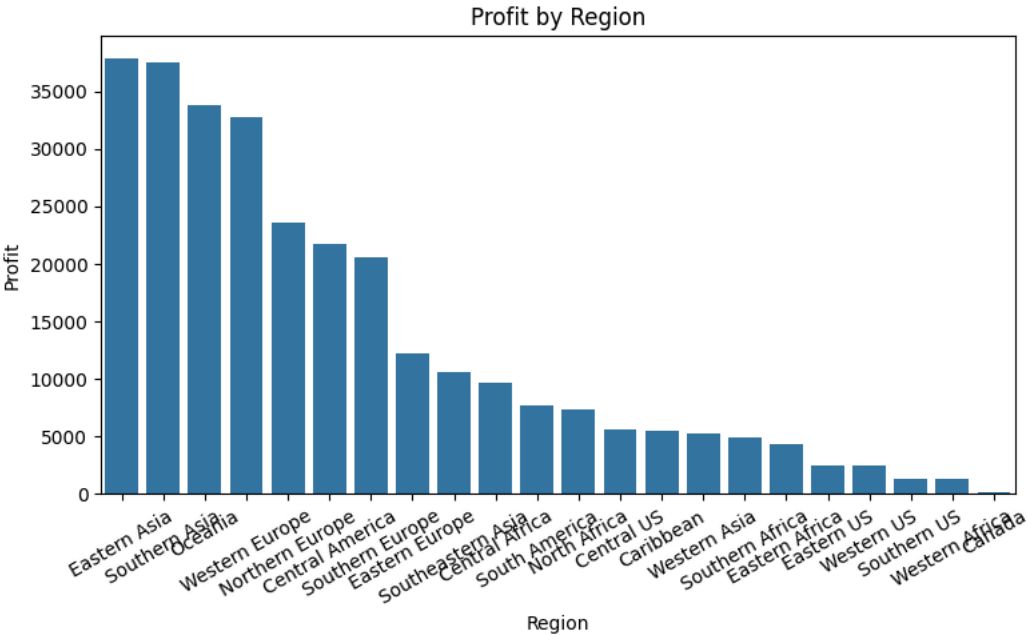
Name: Sales, dtype: float64



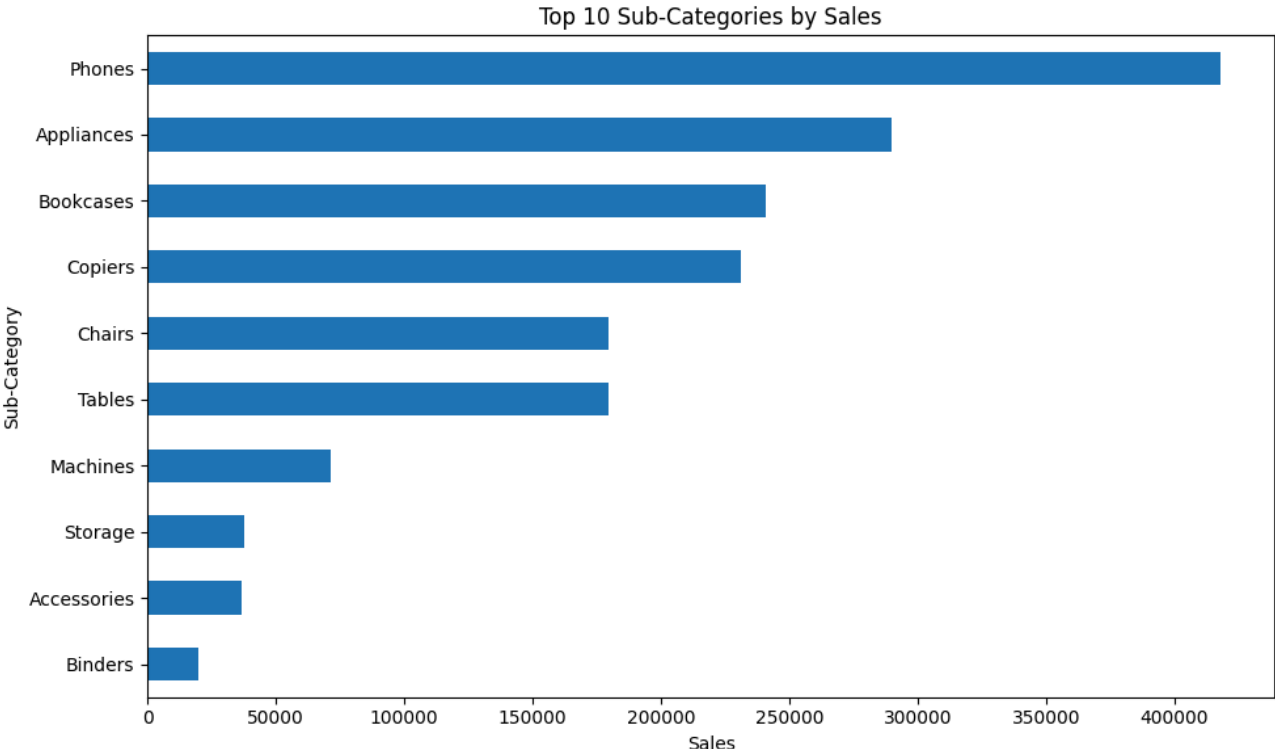
===== PROFIT BY REGION =====

Region	Profit
Eastern Asia	37911.16
Southern Asia	37536.60
Oceania	33777.56
Western Europe	32783.40
Northern Europe	23552.73
Central America	21741.95
Southern Europe	20557.50
Eastern Europe	12228.45
Southeastern Asia	10561.80
Central Africa	9668.22
South America	7723.58
North Africa	7416.42
Central US	5676.02

Caribbean 5463.82
Western Asia 5262.35
Southern Africa 4876.32
Eastern Africa 4302.72
Eastern US 2544.89
Western US 2458.34
Southern US 1384.72
Western Africa 1278.17
Canada 213.72
Name: Profit, dtype: float64

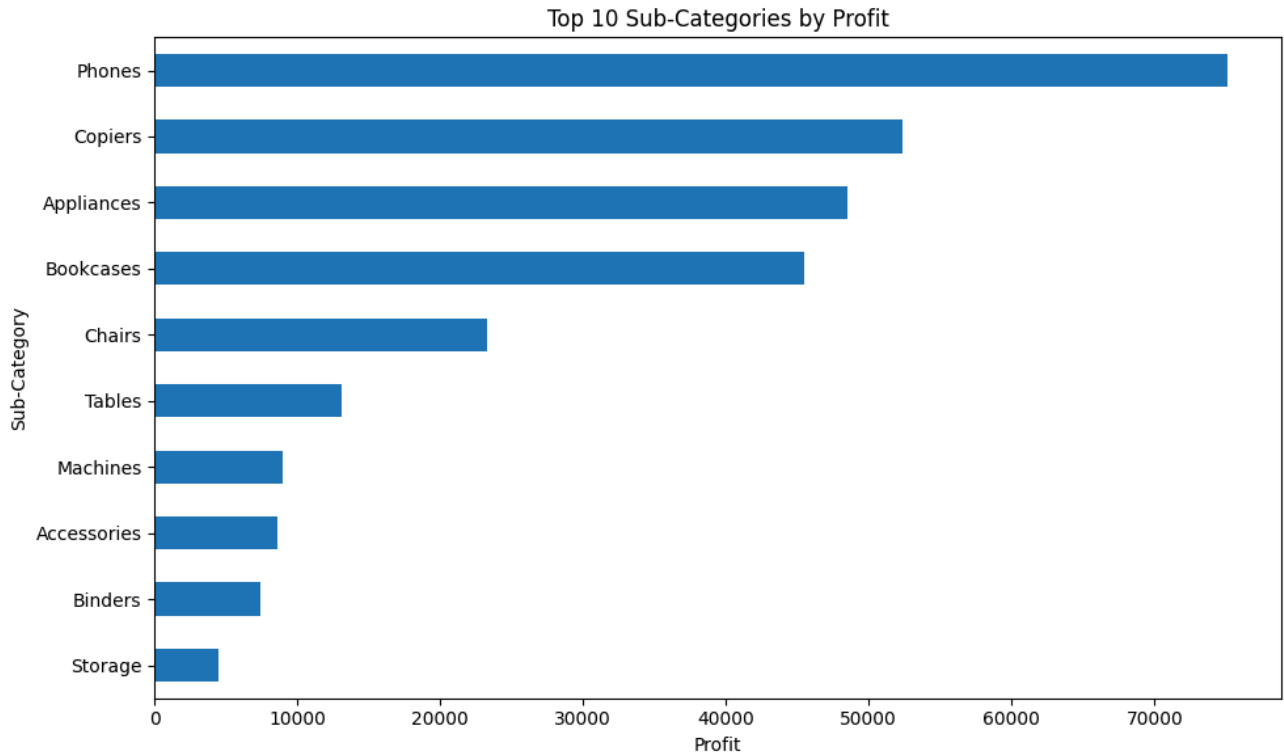


===== TOP SUB-CATEGORIES BY SALES =====
Sub-Category
Phones 417939.31
Appliances 289711.30
Bookcases 240688.73
Copiers 231104.67
Chairs 179648.44
Tables 179292.88
Machines 71366.87
Storage 37592.32
Accessories 36572.17
Binders 19751.19
Name: Sales, dtype: float64



===== TOP SUB-CATEGORIES BY PROFIT =====

```
Sub-Category
Phones      75119.47
Copiers     52315.19
Appliances  48518.99
Bookcases   45483.14
Chairs      23245.88
Tables      13051.69
Machines    8992.06
Accessories 8641.96
Binders     7435.09
Storage     4490.04
Name: Profit, dtype: float64
```



===== LOSS-MAKING SUB-CATEGORIES =====

```
Sub-Category
Supplies    -89.85
Name: Profit, dtype: float64
```

===== CORRELATION =====

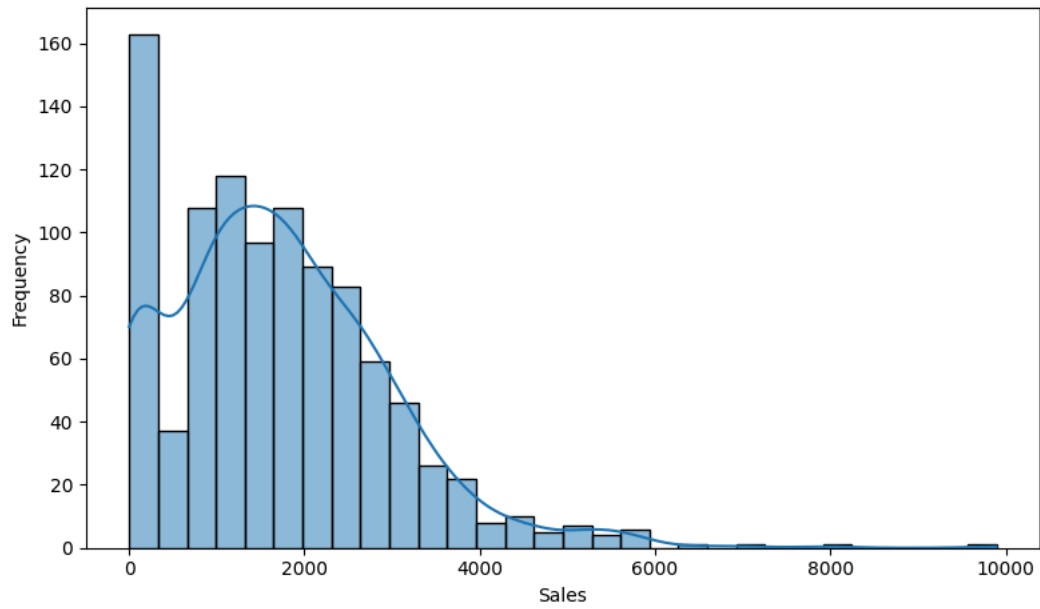
```
      Sales  Profit
Sales  1.000000  0.533676
Profit  0.533676  1.000000
```



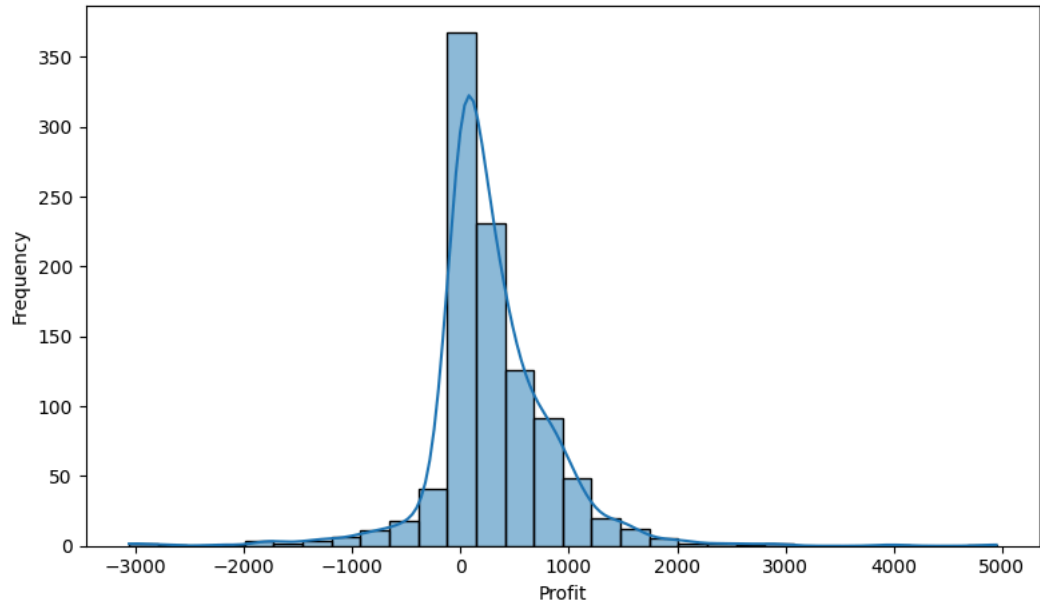
Sales

Profit

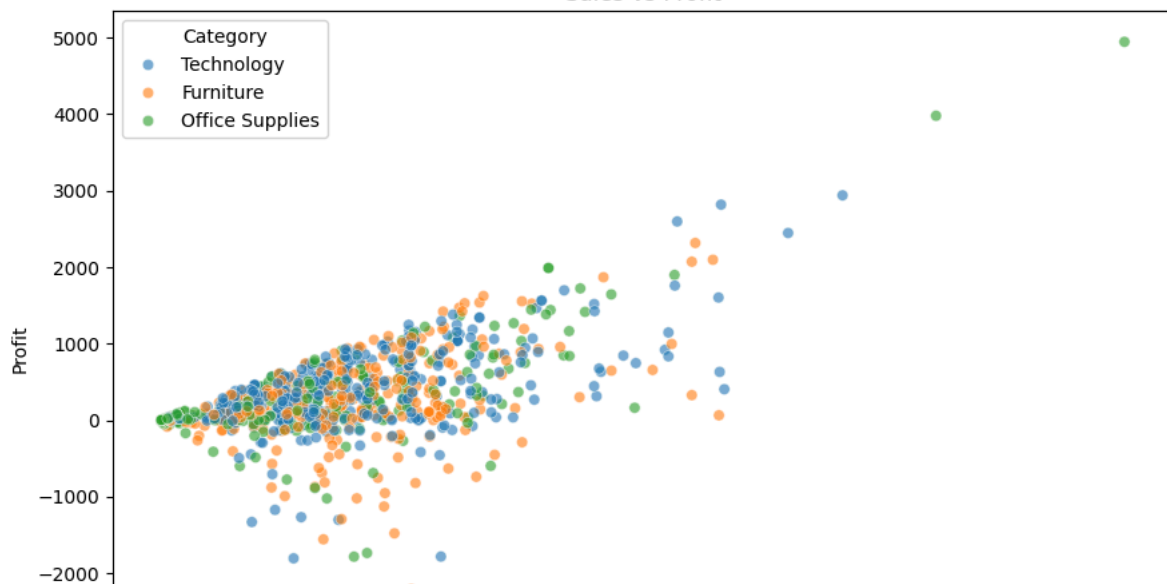
Sales Distribution

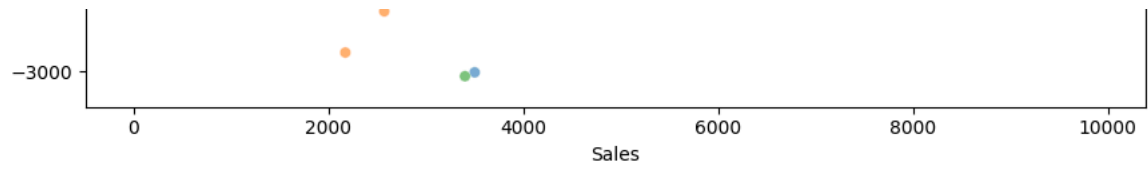


Profit Distribution



Sales vs Profit





Interactive Sales by Category

